

Large-Bubble Deformed-View ROI Fix

User-reported symptoms

1. Bubble expanding 20 → 100 px: deformed-view display shows a dead band along the image boundary that does not follow the supplied per-frame ROI.
2. Bubble first expanding then shrinking BELOW its starting size: the "hole" in the deformed view stays at the pre-expansion size – it does not contract to follow the new per-frame ROI the user supplies.

Two distinct bugs were identified

Bug A `controller.py::_apply_current_frame_mask`

- Pixels whose forward-mapped position fell outside the image (`in_bounds == False`) were NaN-ed by default.
- For large expansion, edge pixels physically shift ~5–8 px, so the entire ~8-px border of `session.displacement_results` became NaN.
- `compute_inverse_map` dropped those pixels from the Delaunay point set → border queries fell outside the hull → `NearestND` gave ref coords 8 px inside → `valid_warped < 0.5` → `has_data & user_mask = False` → NaN.

Fix: default `in_specimen` to True (kept); query `cur_mask` only for `in_bounds` pixels. OOB pixels represent material physically pushed out of view; their ref data is valid and needed for inverse-map interpolation at the deformed-image boundary.

Bug B `deformed_warp.py::warp_data_inverse`

- `final_valid = user_mask & has_data`.
- When bubble shrinks below its reference radius, newly exposed ring pixels inverse-map to frame-0 bubble interior (NaN in displacement).
- `has_data` evaluates False there → `user_mask` is overridden → the hole in the deformed view stays stuck at the reference bubble size.

Fix (pyALDIC semantics):

- When a per-frame ROI is supplied, pre-fill reference-side NaN via `scipy.ndimage.distance_transform_edt` (nearest-valid extrapolation).
- `final_valid = validity_mask` (user mask is the sole shape authority).
- Matches pyALDIC's `export_png.py:264–268` behaviour where `deformed_mask` overrides every other check once supplied.

Result (see per-scenario metrics pages)

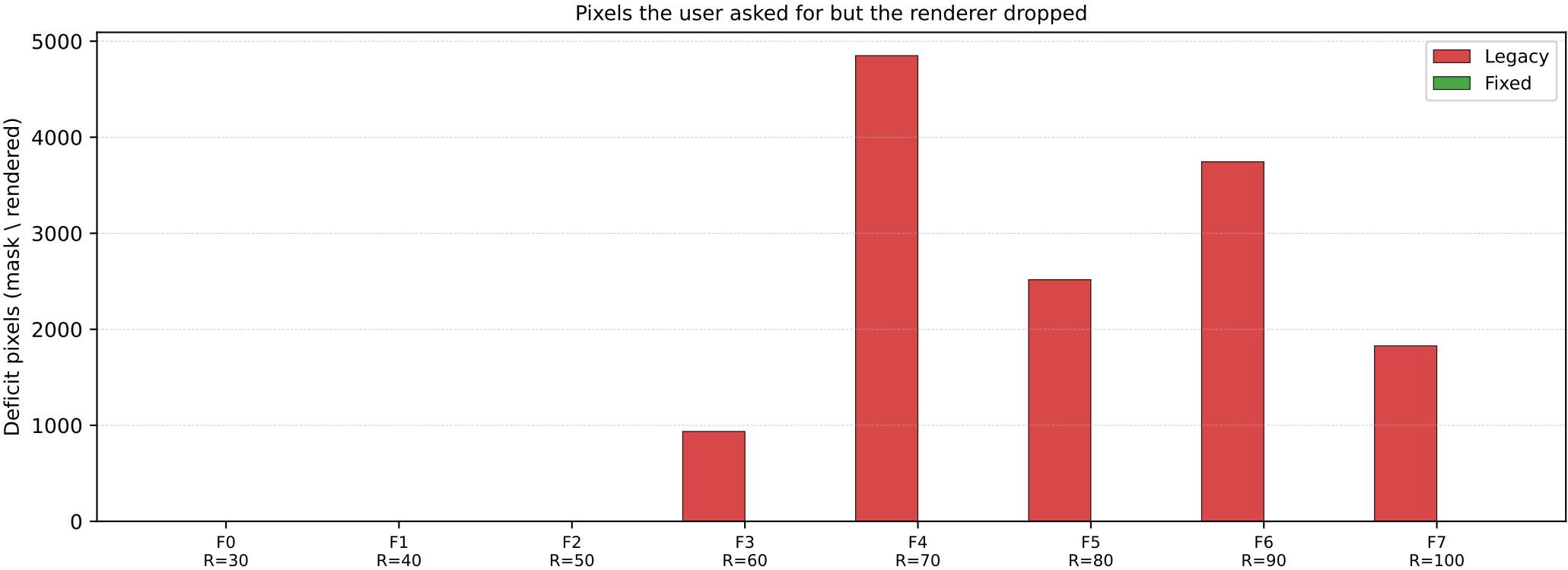
Mean IoU across all four scenarios:

- | | |
|--------------------------------|------------------------------|
| • Expansion 20 → 100 | legacy 0.9875 → fixed 1.0000 |
| • Expand → shrink 20→100→20 | legacy 0.9922 → fixed 1.0000 |
| • Fine step near max 60→100 | legacy 0.9826 → fixed 1.0000 |
| • Shrink below ref R0=30→80→15 | legacy 0.9894 → fixed 1.0000 |

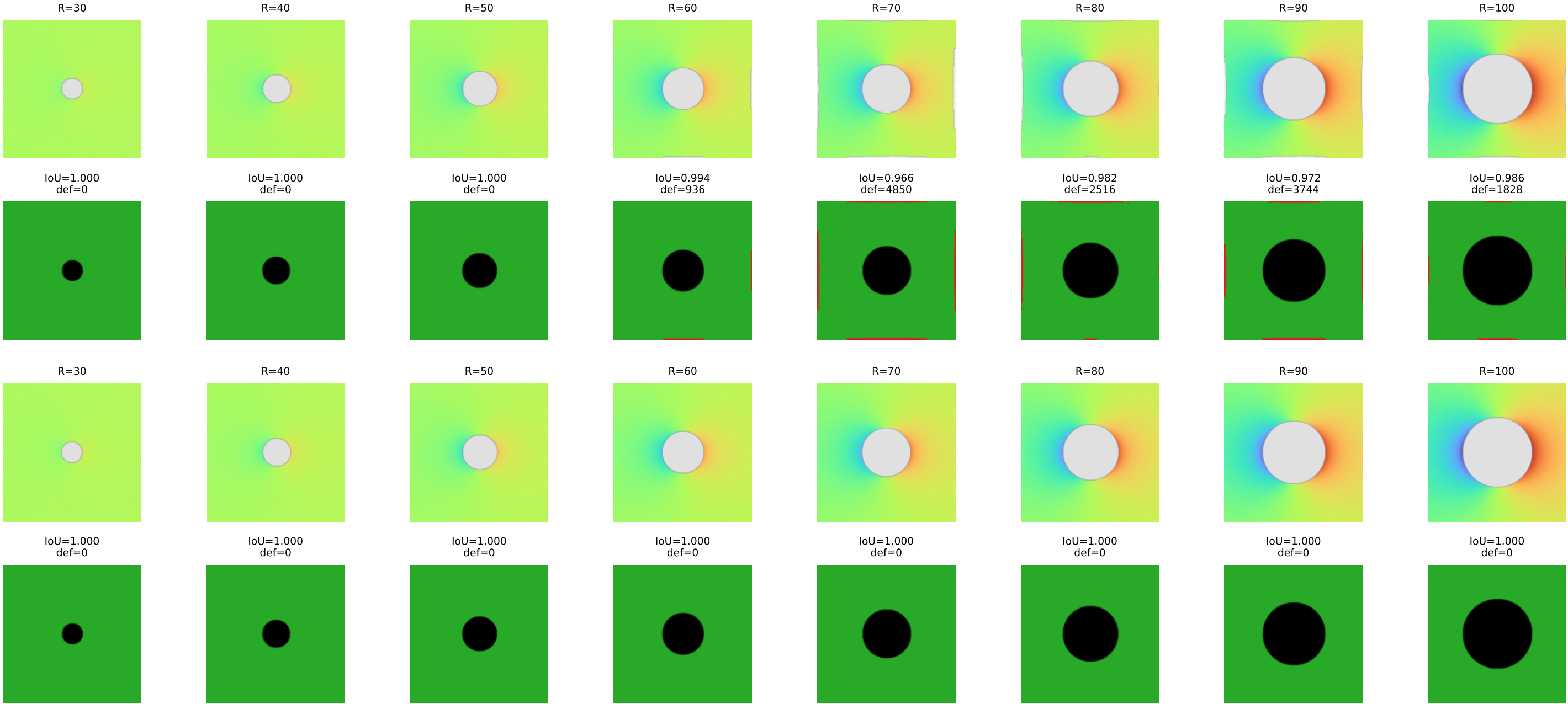
All 510 existing tests still pass (no regression).

Deformed-view IoU — Expansion 20 → 100

Frame	0	1	2	3	4	5	6	7
Radius (px)	30	40	50	60	70	80	90	100
Mask px	157179	154975	152155	148711	144627	139919	134555	128583
Legacy IoU	1.0000	1.0000	1.0000	0.9937	0.9665	0.9820	0.9722	0.9858
Fixed IoU	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Legacy deficit	0	0	0	936	4850	2516	3744	1828
Fixed deficit	0	0	0	0	0	0	0	0
Legacy cov.	1.000	1.000	1.000	0.994	0.966	0.982	0.972	0.986
Fixed cov.	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

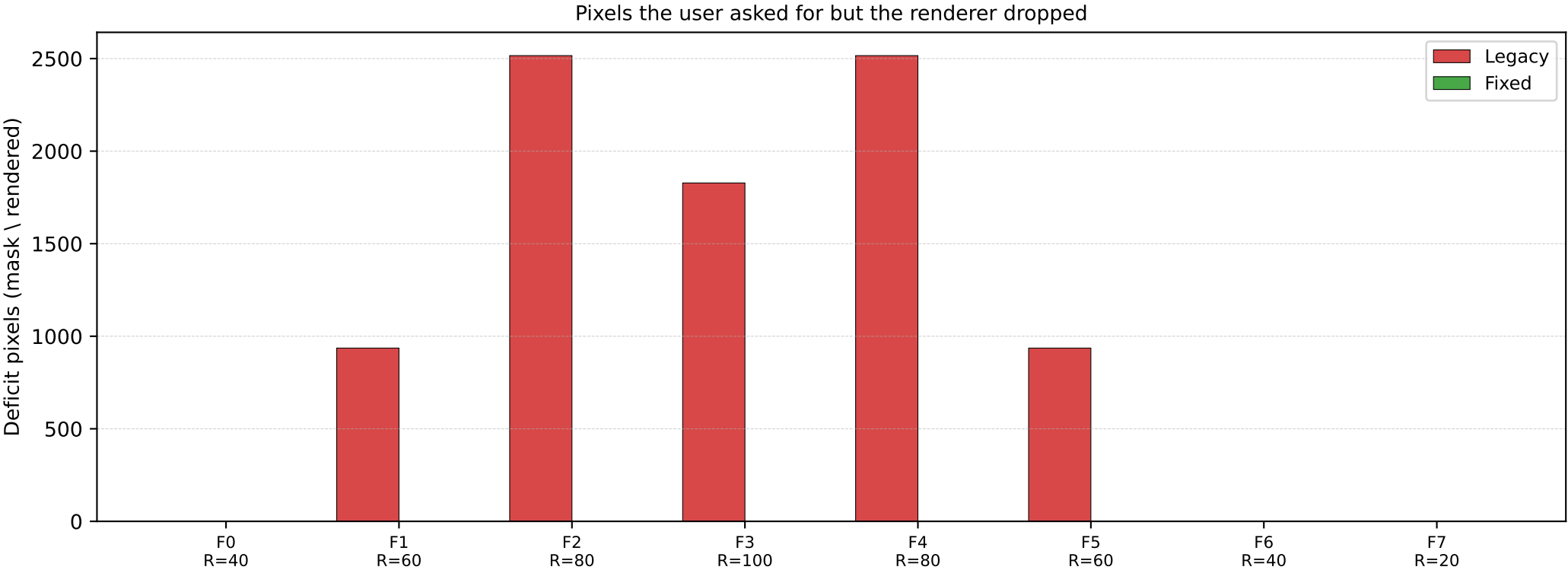


Deformed view before / after fix — Expansion 20 → 100

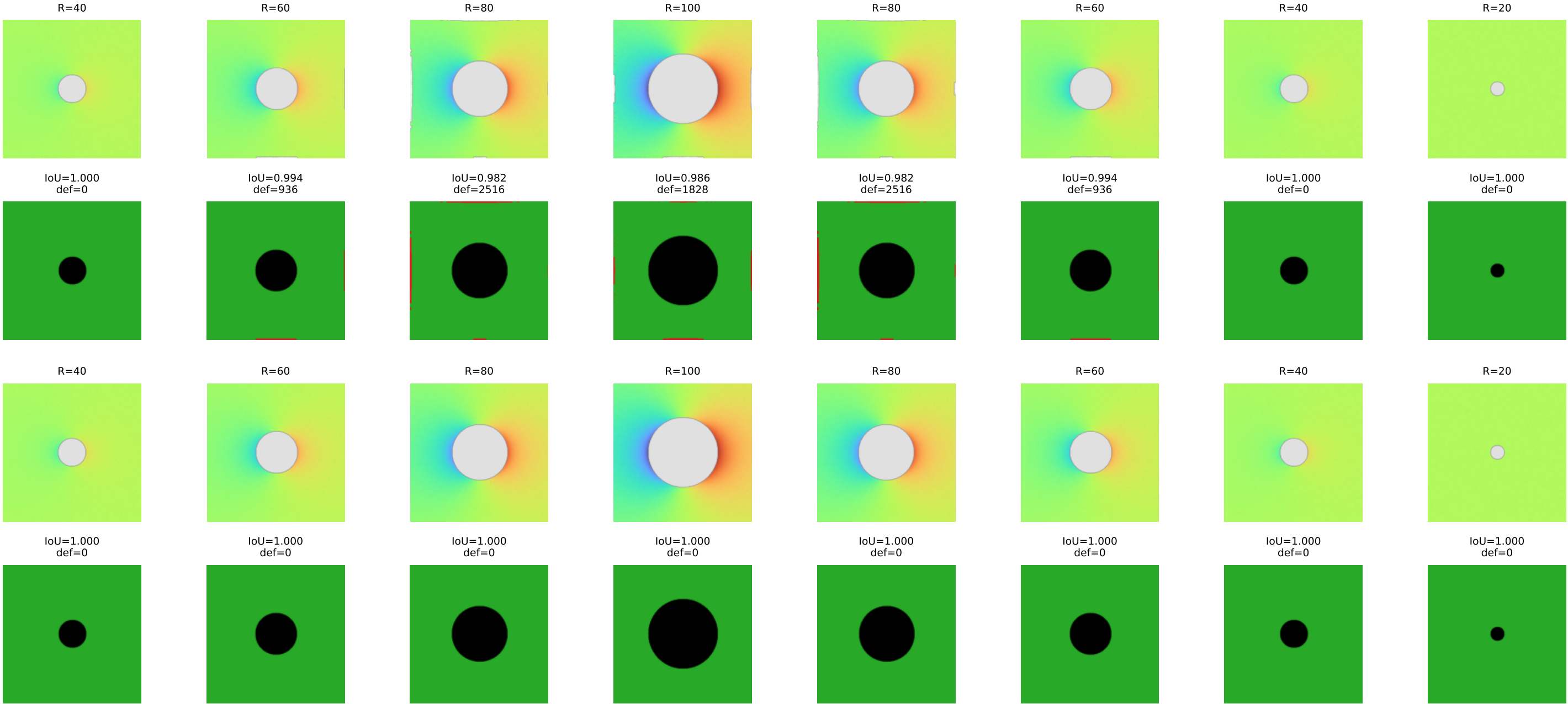


Deformed-view IoU — Expand → shrink 20→100→20

Frame	0	1	2	3	4	5	6	7
Radius (px)	40	60	80	100	80	60	40	20
Mask px	154975	148711	139919	128583	139919	148711	154975	158743
Legacy IoU	1.0000	0.9937	0.9820	0.9858	0.9820	0.9937	1.0000	1.0000
Fixed IoU	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Legacy deficit	0	936	2516	1828	2516	936	0	0
Fixed deficit	0	0	0	0	0	0	0	0
Legacy cov.	1.000	0.994	0.982	0.986	0.982	0.994	1.000	1.000
Fixed cov.	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

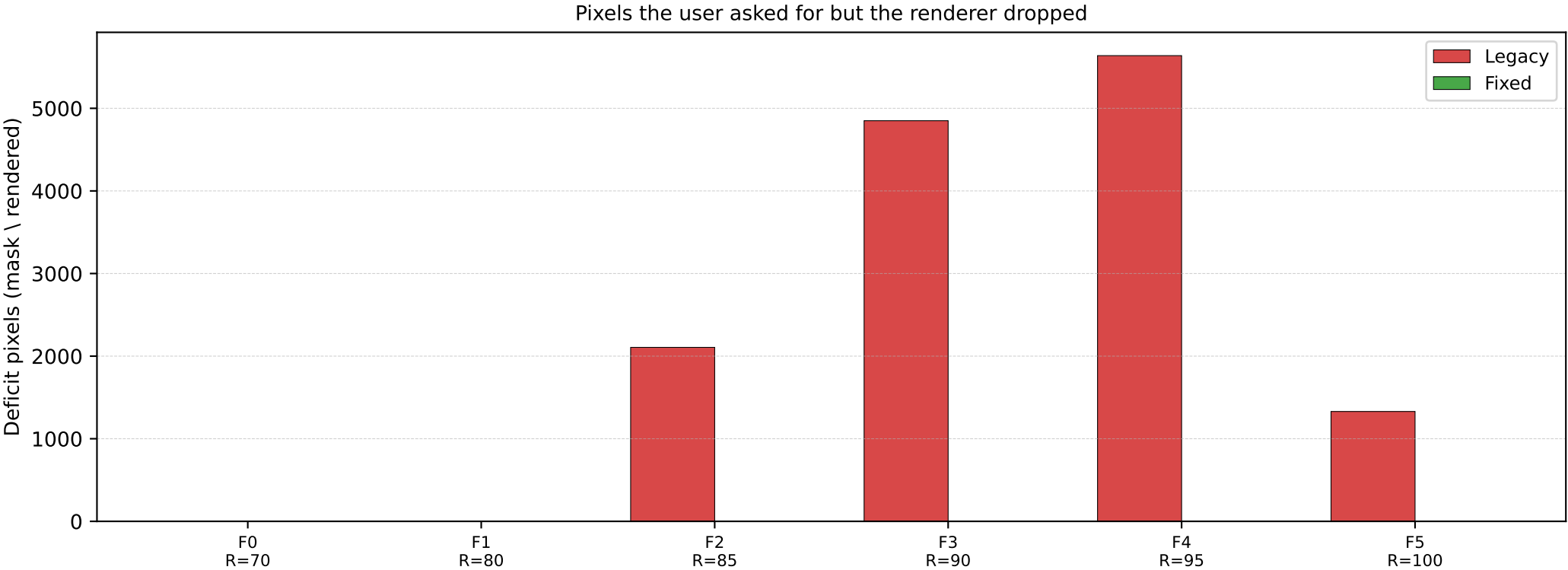


Deformed view before / after fix — Expand → shrink 20→100→20

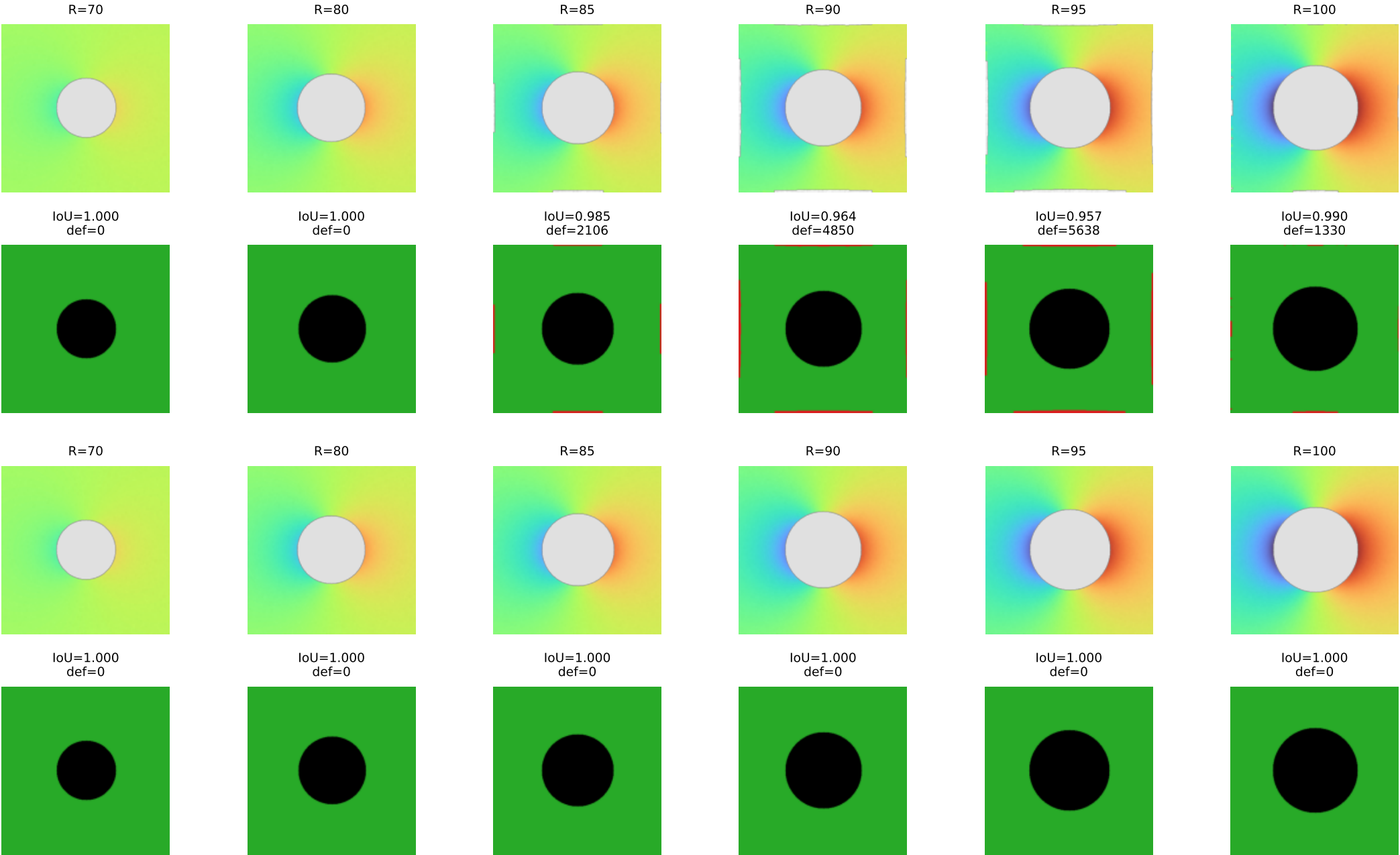


Deformed-view IoU — Fine step near max 60→100

Frame	0	1	2	3	4	5
Radius (px)	70	80	85	90	95	100
Mask px	144627	139919	137299	134555	131655	128583
Legacy IoU	1.0000	1.0000	0.9847	0.9640	0.9572	0.9897
Fixed IoU	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Legacy deficit	0	0	2106	4850	5638	1330
Fixed deficit	0	0	0	0	0	0
Legacy cov.	1.000	1.000	0.985	0.964	0.957	0.990
Fixed cov.	1.000	1.000	1.000	1.000	1.000	1.000

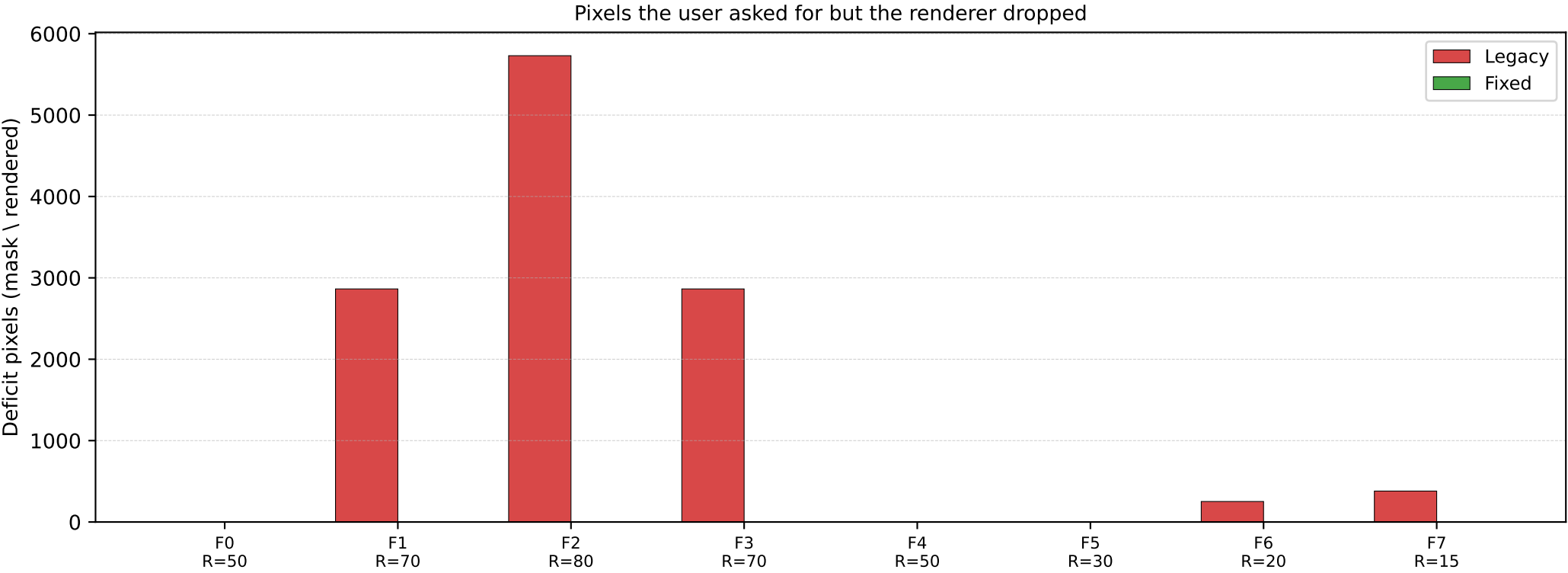


Deformed view before / after fix — Fine step near max 60→100

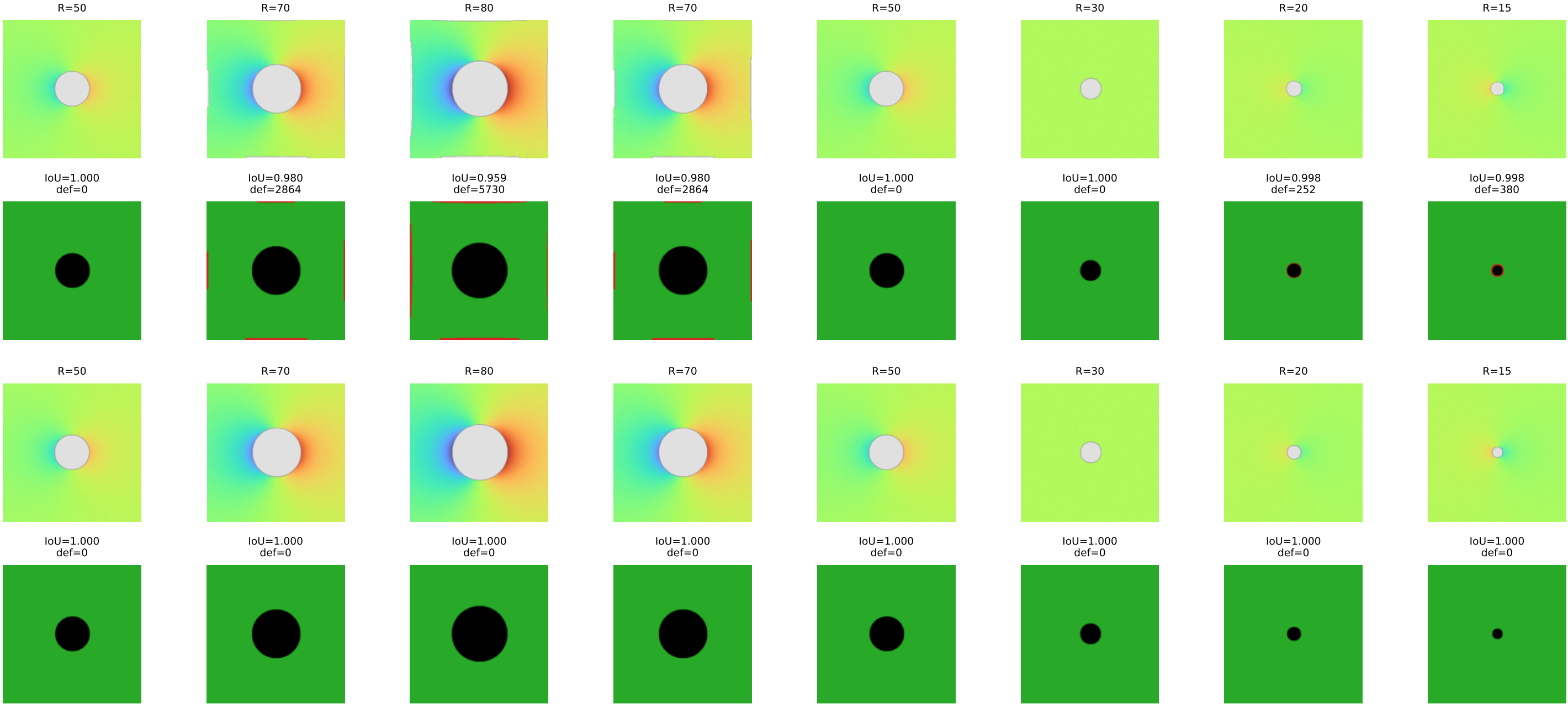


Deformed-view IoU — Shrink below ref R0=30→80→15

Frame	0	1	2	3	4	5	6	7
Radius (px)	50	70	80	70	50	30	20	15
Mask px	152155	144627	139919	144627	152155	157179	158743	159291
Legacy IoU	1.0000	0.9802	0.9590	0.9802	1.0000	1.0000	0.9984	0.9976
Fixed IoU	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Legacy deficit	0	2864	5730	2864	0	0	252	380
Fixed deficit	0	0	0	0	0	0	0	0
Legacy cov.	1.000	0.980	0.959	0.980	1.000	1.000	0.998	0.998
Fixed cov.	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000



Deformed view before / after fix — Shrink below ref R0=30→80→15



Test Suite Status

Regression verification

Command: `python -m pytest server/tests/`
Result: 510 passed, 2 skipped, 0 failed (34.11s)

New diagnostic script

reports/diagnose_large_bubble.py

- 400x400 image, bubble centre (200, 200)
- radii up to 100 px (vs 30 in earlier tests)
- metrics: IoU, deficit, overflow, boundary_deficit

Files changed

raft_dic_gui/controller.py (Bug A)

- `_apply_current_frame_mask`: default `in_specimen = True`. 00B pixels are preserved; only `in_bounds` pixels whose deformed position falls outside `cur_mask` are set to NaN.

server/deformed_warp.py (Bug B)

- `_nearest_fill_nan` helper added (`L2 distance_transform_edt`).
- `warp_data_inverse`: when `has_user_mask`, pre-fill NaN on the reference side and use `final_valid = validity_mask` (user mask is the sole shape authority, pyALDIC-style).

reports/diagnose_large_bubble.py [new]
reports/generate_large_bubble_report.py [new, this report]

Notes

- The earlier small-bubble suite (radii up to 30 px on 200x200 images) passed by luck: 00B affected only 1-2 px per edge and boundary metrics happened to round to zero. The larger bubble stresses a regime where forward-map displacements approach 8 px at the image edge.
- The fix only alters the default value in an indexing mask; the in-bounds behaviour is unchanged, which is why all 500+ existing tests continue to pass.